

Case Study Scope of Work

Project Title: AI Job Market Intelligence: Analyzing Demand for AI Skills in Europe

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Stakeholders: Students, Universities, Career Advisors, Recruiters

Date: June 2026

Purpose

The goal of this project is to analyze publicly available AI and Data Science job postings across Europe to identify the most in-demand skills, top roles, geographic hiring trends, salary patterns, and entry-level opportunities. The findings will help students, universities, and career advisors make data-informed decisions about skill development and career planning.

Scope / Major Project Activities

Activity	Description
Data Collection	Source publicly available AI/Data Science job datasets. Document all sources, versions, and access dates.
Data Cleaning	Handle missing values, remove duplicates, standardize job titles and locations, extract and categorize skills from job descriptions.
Data Analysis	Analyze skills demand, role distribution, geographic hiring trends, salary patterns, entry-level opportunities, and skill co-occurrence.
Visualization	Build a Power BI dashboard and supporting charts covering top skills, locations, salaries, and entry-level breakdowns.
Insights & Recommendations	Summarize findings and propose a prioritized skill roadmap for students and actionable guidance for universities and recruiters.

Out of Scope

- Predictive modeling — no salary forecasting or hiring demand prediction.
- Real-time job scraping — dataset-based analysis only, no live LinkedIn monitoring.
- Resume matching or candidate scoring systems.
- LLM applications such as chatbots, RAG systems, or AI agents.

Deliverables

Deliverable	Description / Details
Clean Dataset	Fully prepared dataset with data dictionary and source documentation.
Python Analysis Notebook	End-to-end cleaning and EDA workflow in Jupyter / Google Colab.
Power BI Dashboard	Interactive dashboard for stakeholders — skills, locations, salaries, entry-level.
Final Presentation	10–12 slides: business problem, methodology, findings, and recommendations.
Portfolio Case Study	Professional write-up published on GitHub and LinkedIn.

Success Criteria

The project will be considered successful if it can answer:

#	Question
1	Which AI skills are most in demand across European job postings?
2	Which AI roles have the highest posting volume?

#	Question
3	Which European cities and countries offer the most opportunities?
4	Which skills are associated with higher advertised salaries?
5	What skills should students prioritize to improve employability?

Methodology

The project follows the Google Data Analytics six-phase framework:

Phase	Stage	Key Activities
Phase 1	Ask	Define business questions, identify stakeholders, establish success criteria.
Phase 2	Prepare	Source and document datasets, assess quality, set up project structure.
Phase 3	Process	Clean data in Python — handle nulls, remove duplicates, standardize fields, extract skills.
Phase 4	Analyze	EDA in Python and SQL — skills ranking, salary analysis, geo analysis, skill co-occurrence.
Phase 5	Share	Build Power BI dashboard, prepare final presentation, write case study narrative.
Phase 6	Act	Publish to GitHub and LinkedIn, formulate recommendations, document limitations.

Schedule Overview

Milestone	Target Date	Description / Details
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Data Collection & Cleaning	Week 1	Dataset sourced, cleaned, and documented. Data dictionary complete.
Exploratory Data Analysis	Week 2	All five business questions analyzed in Python and SQL.
Power BI Dashboard	Week 3	Interactive dashboard built and validated with stakeholder scenarios.
Final Presentation & Case Study	Week 4	Slide deck complete. Case study published on GitHub and LinkedIn.

Estimated Completion Date

July 2026